

Atmosic ATM33/e presentation

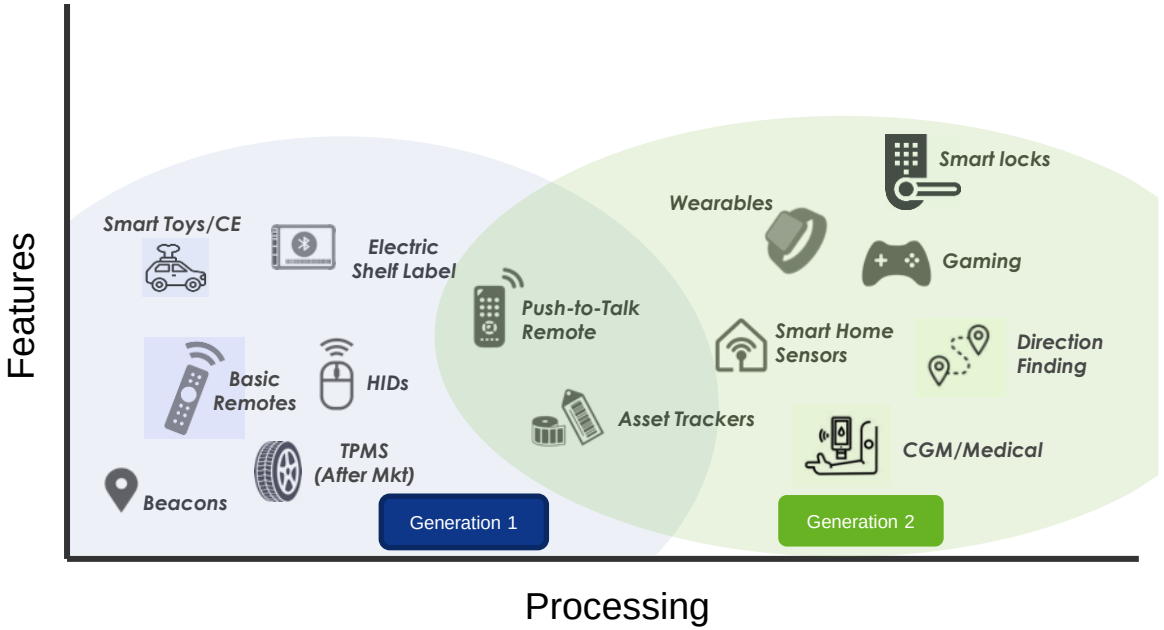
July 2023



SoC Product Roadmap

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The Evolving Low Power IoT Landscape

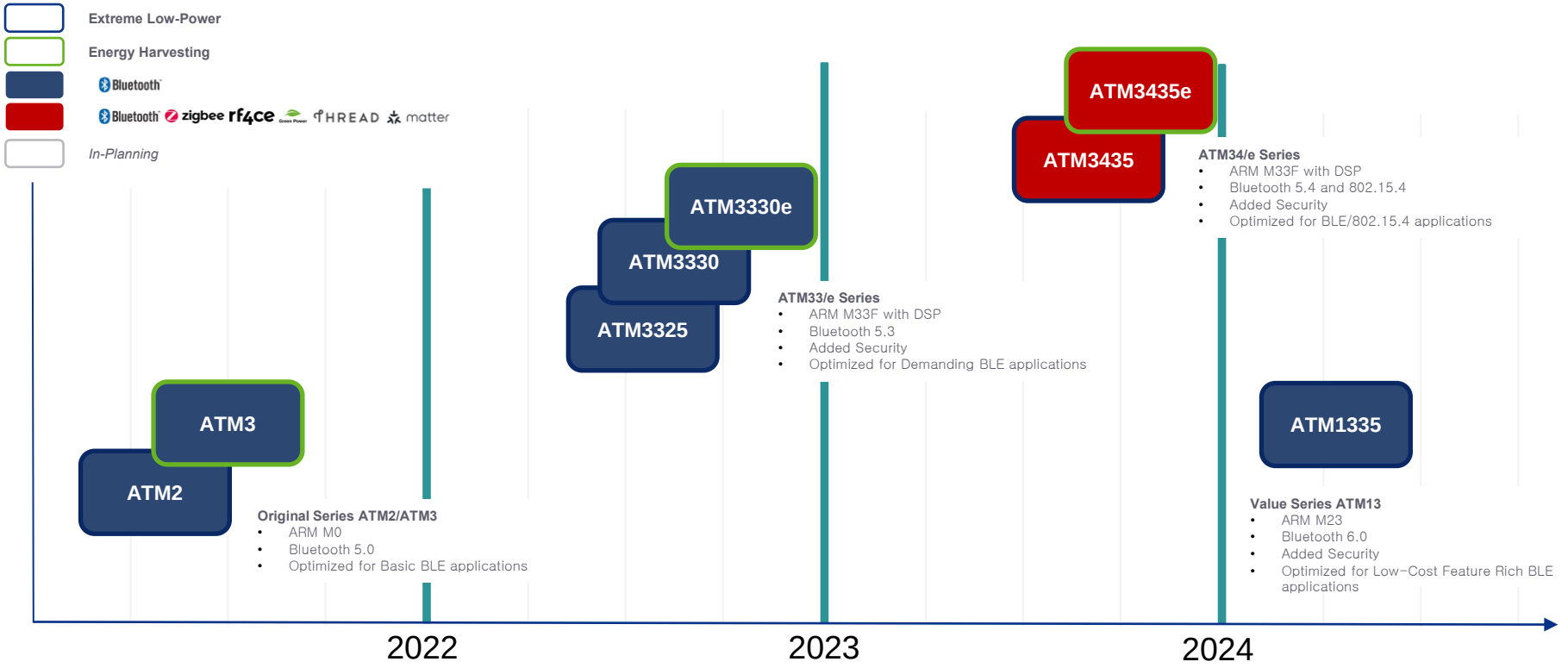


- IoT low-power wireless landscape
  - Billions of Units by 2024
  - Forecast: increased growth in LPW (>10% CAGR 2020-2024)
- Greater demands
  - More processing
  - More features – Security, Locationing
  - Ultra Lower Power
- Atmosic is the only company focusing on developing Ultra Low Power and integrated Energy Harvested solutions

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SoC Roadmap

*Subject to Change



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Product Overview – ATM33/e Series

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Foundation for Product Generations

ATM2/ATM3 Series



+



- More Processing & Embedded NVM
- Security and ARM TrustZone
- Bluetooth LE 5.1/5.2/5.3
- AoA/AoD for Asset Tracking, Direction Finding
- PV/RF Harvester and Battery Support Enhancements
- Sensor Hub Enhancements – MCU-less Operation
- Increased Tx Power

→

ATM33/e Series





Common Foundation for All Product Generations –

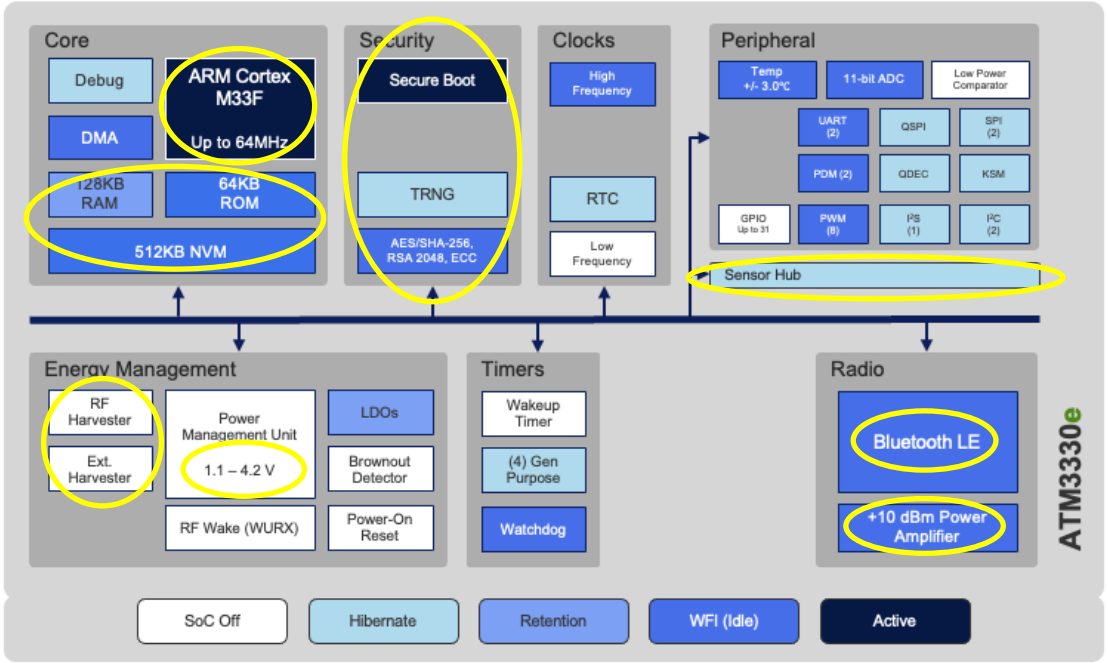
- **Ultra Low Power Consumption**
- **Managed Energy Harvesting (PV, RF, Thermal, Mechanical)**
- **MCU-less Sensing, On-Demand Wakeup Receive (WuRx)**

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ATM33/e Series Product Features

- Bluetooth LE 5.3
- Radio power consumption
  - Receiving: 0.85 mA@-95dBm
  - Transmitting: 2.1 mA@0dBm
- RX Sensitivity: -95dBm
- TX Output: +10dBm
- Energy Harvesting (ATM33e only)
- ARM M33F @ 64MHz
- ARM TrustZone, Secure Boot & Secure Debug, Secure OTA
- TRNG, AES-256 HW Crypto
- NVM 512kB, RAM 128kB, ROM 64kB
- QFN 5x5mm, 7x7mm, WLCSP



ATM3330e - Energy Harvesting
ATM3330 – non Energy Harvesting

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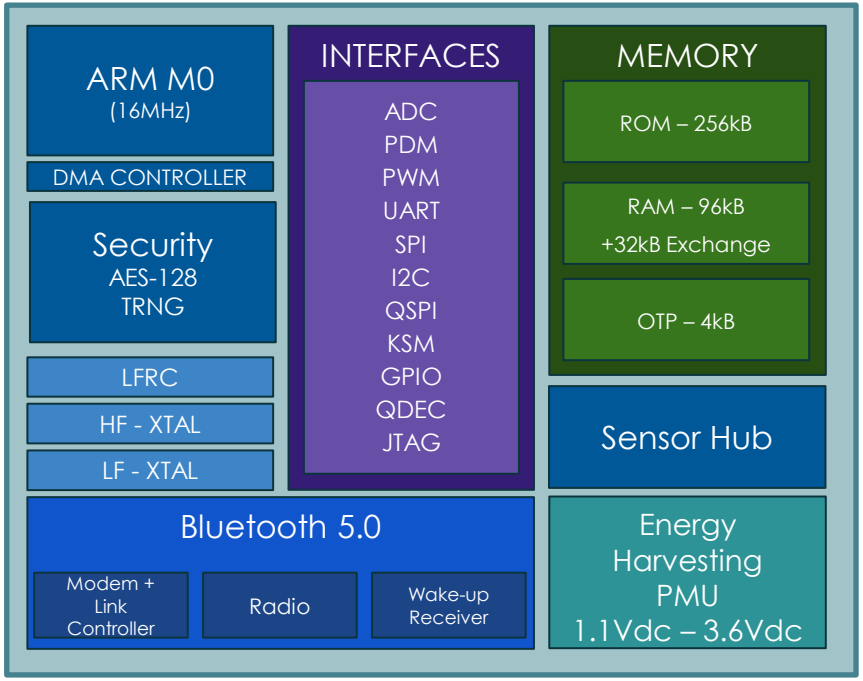
ATM33/e Chip Packages

P/N	Package	Energy Harvesting	Wake Up Receiver	GPIOs
ATM3330e	7x7 QFN, 56 pins	YES	YES	31
ATM3330	7x7 QFN, 56 pins	NO	YES	31
ATM3325	5x5 QFN, 40 pins	NO	NO	21
ATM3325	WLCSP, 49 pins 2.610mm x 2.430mm x 0.377mm	NO	NO	21

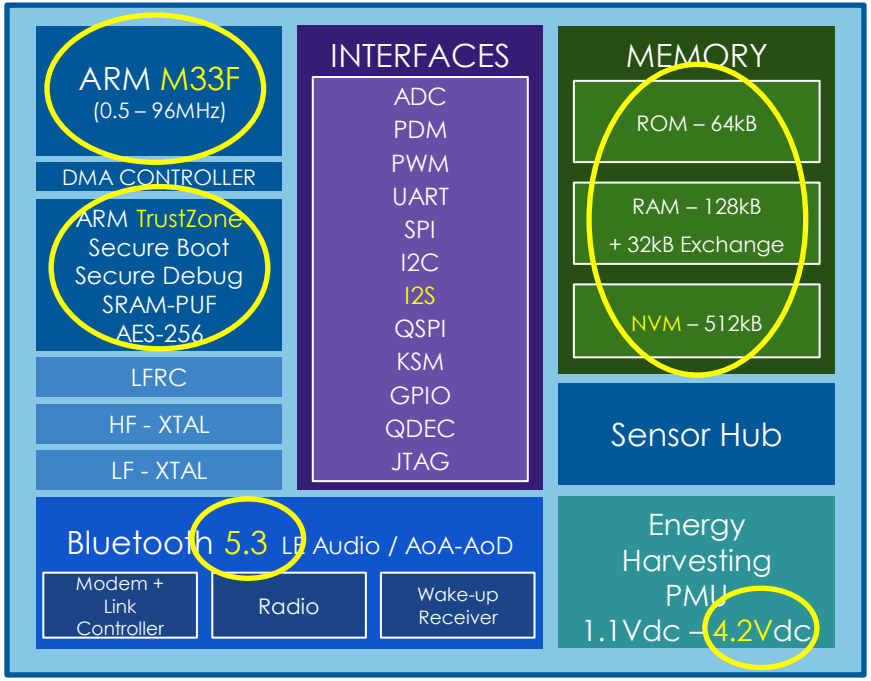
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Gen1(ATM2/3) vs Gen2 (ATM33/e)

ATM2/3



ATM33/e



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Gen 1 (ATM2 / ATM3) & Gen 2 (ATM33/e) Applications

ATM2/ATM3

- HID Applications
  - PTT Remote
  - BT Remote
  - KB
  - Mouse
- ESL/E-badge
- Toys
- Single Sensor
- TPMS (After Market)
- People Tracking
- Re-use, High-value Asset Tracking
- Beacons

ATM33/e

- Medical/Healthcare
  - CGM
  - Medical ID/Wristband
  - Asset/Equipment/Sample/Storage tracking
- Wearables
  - Fitness band
  - Smart rings/Jewelry/Clothing
- HID Applications
  - AO/PTT Remote
- Accurate Location and Direction Finding
- Gaming
- Sensor Fusion
- Home Automation – Smart locks, Switches
- Automotive - Anomaly Detection/Sensor

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Features Overview

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System Features Overview

- **Bluetooth LE 5.1/5.2/5.3**
  - AoA, AoD (5.1)
  - LE Power Control (LEPC)(5.2)
  - Enhanced Attribute Protocol (EATT)
- **Processing Core**
  - ARM M33F - Support CMSIS (Cortex Microcontroller Software Interface Standard)
- **Memory**
  - ROM: 64kB, NVM: 512kB (16kB Cache), RAM: 128kB
  - Memory Protection Unit (MPU), Memory Protection Controller (MPC)
- **Clocks**
  - Xtal oscillator for HF & LF
  - Internal RCO for 32KHz
  - LFRC compensation algorithm to allow stable connections over longer intervals
  - MCU clock supports 64MHz
- **Audio Interfaces**
  - Stereo PDM Rx
  - I2S/PCM Master/Slave, Bi-directional
- **Peripheral Interfaces**
  - 2x I2C (Master), 2x SPI (Master/Slave)
  - Quad SPI for external Flash
  - 11-bit ADC
  - 2x UART, 8x PWM
  - Timers, Power-On-Reset, Watchdog
  - SWD, JTAG
- **Security**
  - ARM Trustzone
  - AES-256, SHA-2 256 Cryptographic Accelerators
  - Secure Boot, Secure Execution, Secure OTA
  - Secure key storage & management
- **Wake-up Mechanisms** (from hibernation/retention)
  - GPIO, KSM/keyboard, QDEC/mouse, SPI, I2C
  - Countdown timer
  - Sensor hub
  - PMU (timer, GPIO level, battery/energy detector, brownout)
  - Wakeup receiver (WuRx)

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System Features Overview

…Continued

- **Radio**
  - 50 Ohm matched single ended RF out
  - Tx Output: +10dBm
  - Interface to support RF FEM
- **Power Management Unit (PMU)**
  - Operating Range: 1.1Vdc – 4.2Vdc
  - Supports Li-Ion and LiFePO4 batteries, hybrid supercaps
  - Rechargeable battery monitoring and protection
- **Power Consumption @ 3V**
  - Radio:
    - *Rx ~0.85mA@-95dBm,*
    - *Tx ~2.1mA@0dBm*
  - Retention, 32kB SRAM: 1.6µA
  - Hibernate with WuRx: 1.4µA
  - Hibernate: 1.1 µA
  - SoC off: 400 nA
  - PowerDown: 100 nA

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Software Features

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Software Features and Tools

- **Platforms Supported**
  - Windows10, 11
  - Linux Ubuntu LTS
  - MacOS
- **Compiler**
  - gcc
  - Support for Keil ARM C/C++, IAR C/C++ compilers
- **IDE Platforms**
  - IAR Workbench, Keil MDK, Segger Embedded Studio
- **RTOS**
  - Zephyr v3.x, FreeRTOS(Planned)
- **Debug Tools**
  - SWD, J-Link/J-link OB, OpenOCD
- **Peripheral Drivers**
  - QSPI, SPI, UART, I2C, PWM, PDM, I2S
- **Application Development Framework**
  - Unified framework for developing applications across all platforms
  - GAP, GATT Profiles
- **Security**
  - ARMv8-M Trustzone
  - Secure Boot, Execution & debug
  - Secure FW Update (OTA)
  - Secure Key Storage & Management
  - Crypto HW Acceleration: AES-256, SHA2 256, HMAC
  - Device Life-cycle management
  - PSA Level1 Certification
- **Application Examples, Services and Profiles**
  - HID Remotes, Keyboards
  - RTLS Tags, Sensor Hub
  - Health Monitors
  - Advanced Wearables & Trackers
- **Mobile Application (Android & iOS)**
  - Scanner, Configuration(AT Command)
  - OTA Support
- **Documentation**
  - SDK User Guides
  - Application Framework Development Guide
  - API Reference Document

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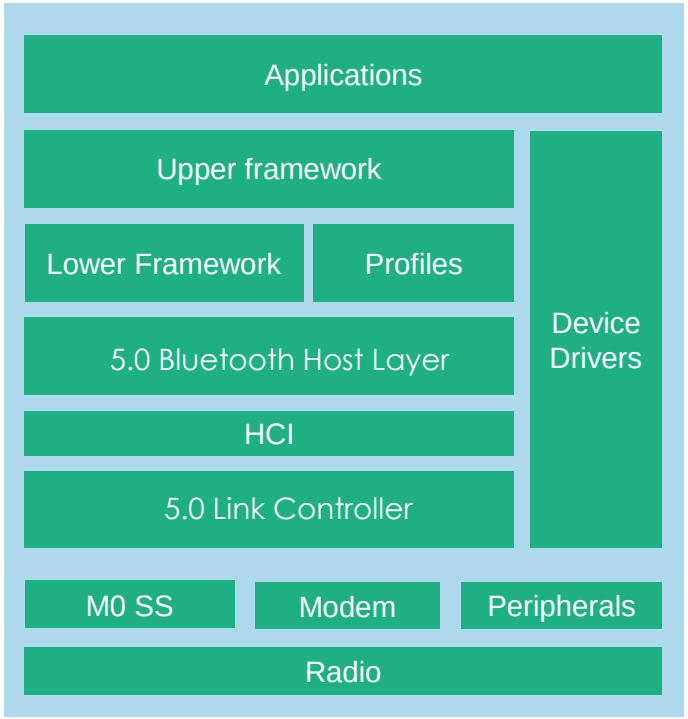
Security Software Features

- Trusted Firmware-M (TFM) v1.4
  - Secure Processing Env for ARMv8-M for secure services
  - Secure Platform Root of Trust
- PSA Certification (Platform Security Architecture)
  - IoT Security Framework defined by ARM
  - Level 1 with Functional API certification
- Zephyr as the baseline RTOS for certification
  - FreeRTOS (Planned)
- MCUBoot
  - Secure Boot, FWUpdate OTA
  - Secure Debug
- Full Security Life Cycle Management
  - Manufacturing → Field Use → RMA → Decommissioning

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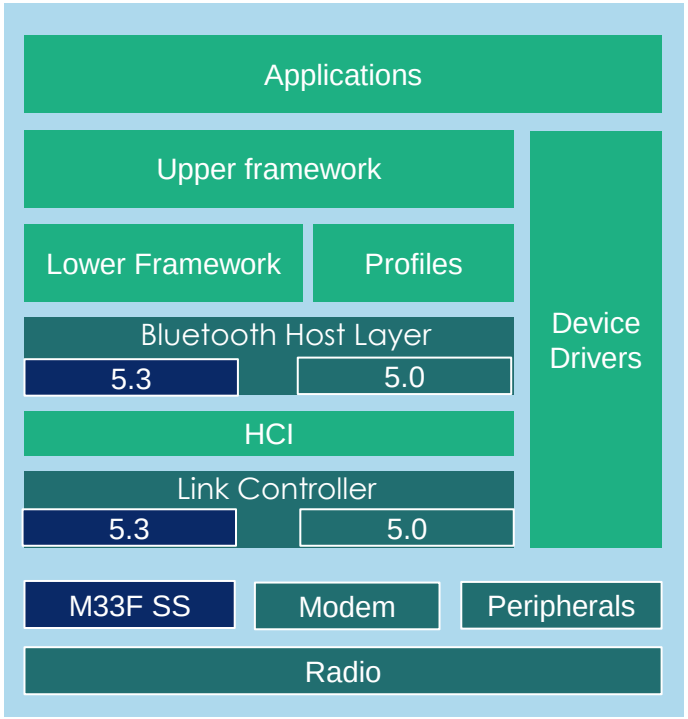
Unified Software Architecture

Gen1: ATM2/ATM3



- Common code base across Gen1 & Gen2
- User Applications remain consistent between the **two** generations
- Hardware peripherals are identical
- Common Tools and File Structure

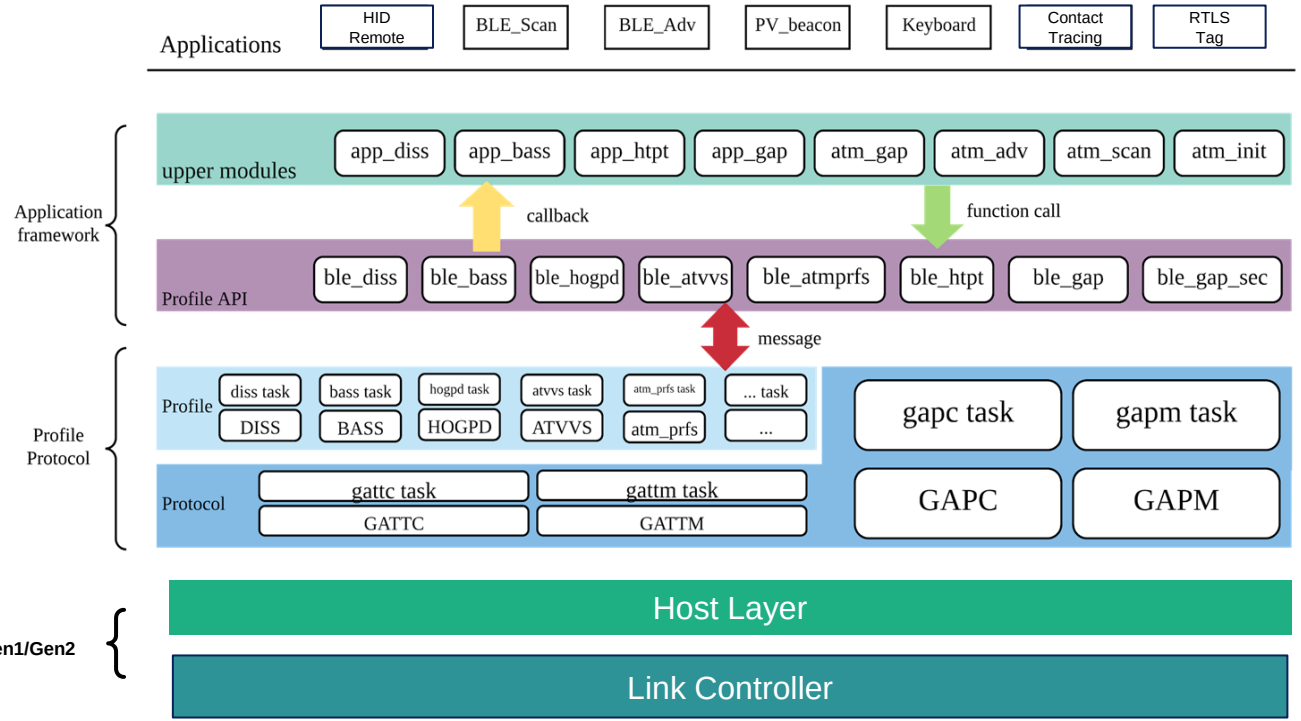
Gen 2: ATM33/e



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Application Development Framework

- Unified Application Development
  - Single application for both platforms
  - Easy to migrate across device families
  - Enables fast prototyping & development
- C library functions built on Bluetooth LE Stack from CEVA
- Majority of Bluetooth LE Profiles supported
- Simplifies porting from other BLE solutions



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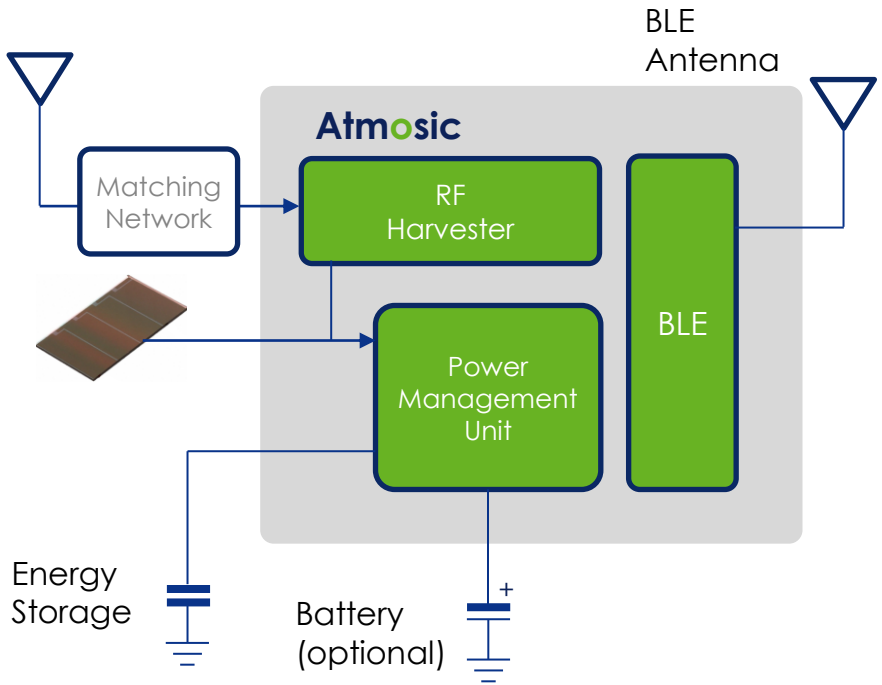
Energy Harvesting and PMU

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Integrated Energy Harvesting & Power Management Unit

- Power Management Unit Manages Multiple Energy Inputs
- On-chip RF Harvester can be tuned to frequencies between 400 MHz and 2.5 GHz
- Separate input for harvested energy from external sources (PV, Thermal, Mechanical)
- Excess harvested energy storage can be a capacitor, supercapacitor, or rechargeable battery
- Supports standard or rechargeable battery configurations
- PMU prioritizes energy requests across available sources to maximize battery life :

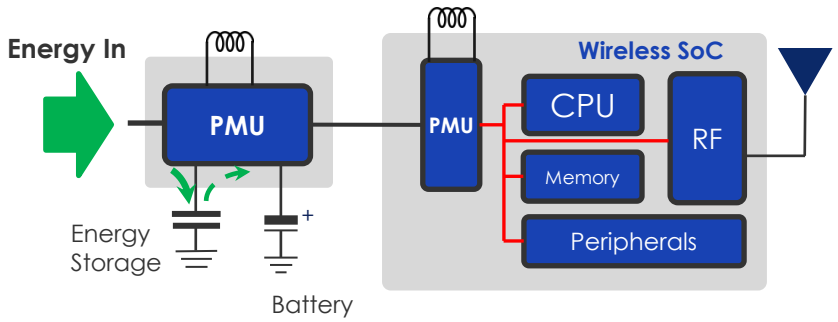
Harvested → Stored → Battery



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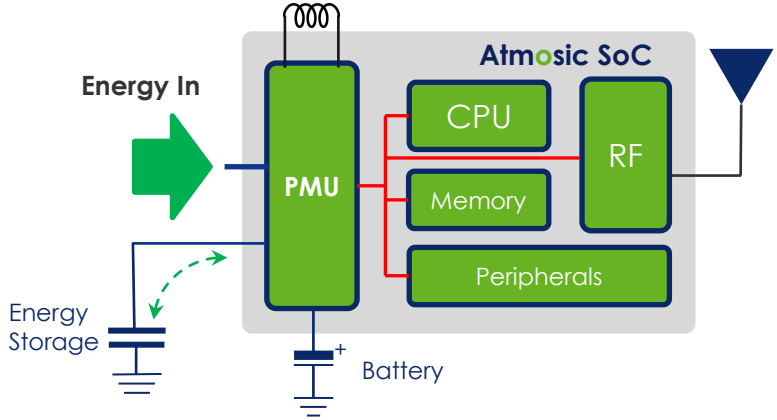
Atmosic vs. Discrete Energy Harvesting PMU

Traditional Discrete PMU Solution



- Discrete solution duplicates PMU functions across two devices
  - Increases BOM
  - Decreased efficiency
  - Reduced flexibility

Atmosic Integrated Solution

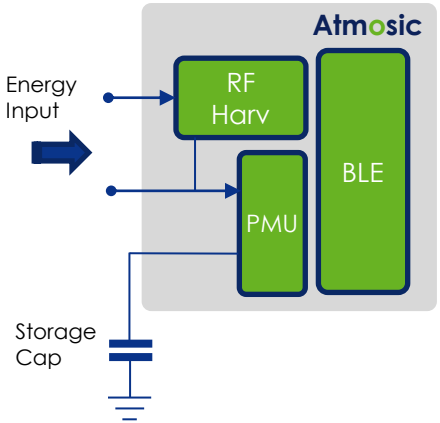


- Integrated solution benefits
  - Lower Overall BOM
  - Direct use of harvested energy
  - CPU controlled configurability and monitoring

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Energy Storage Configurations

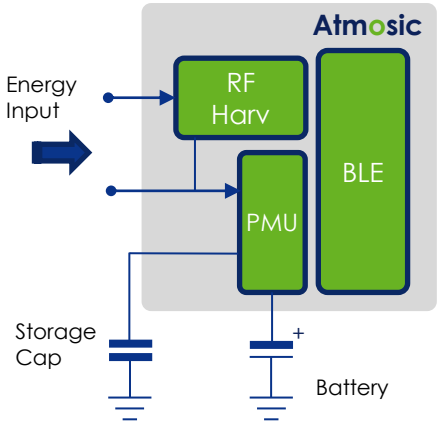
Battery Free



- Capacitor or Supercap for temporary storage of harvested energy
- Hybrid supercap supports long term battery free storage



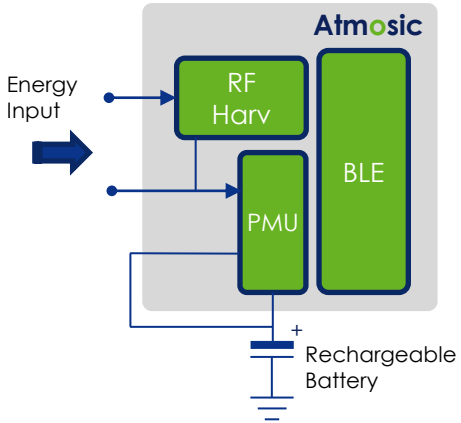
Standard Battery



- Extend battery life with harvesting
- Harvested and stored energy used before battery



Rechargeable Battery



- Excess harvested energy recharges battery
- PMU manages battery charging



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Changing the Power Equation

EXTREME LOW-POWER		ENERGY HARVESTING		ENERGY STORAGE		YOUR PRODUCT
 Bluetooth®	+	Motion, RF, Thermal, Light	+	Capacitor	=	**Battery Free**
			+	Standard Battery	=	**Extended Battery Life**
			+	Rechargeable Battery	=	**Batteries Last Forever**

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Energy Harvesting and Atmosic BLE

RF

1/s beacon up to 5m/2m
from 900MHz/2.4GHz
36dBm source

PV

5-7cm^2 PV @200lux
sufficient for 1/s beacon

One button press
generates 9-12 beacons

Mechanical

5°C Temperature gradient
supports 1/s beacon

Thermal

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Competition

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Atmosic ATM33 vs. Competition

	Atmosic ATM33	NRF52840	NRF52833	NRF52832	TLSR8271	EFR32BG22
Radio-only RX Power (@3V)	**0.85 mA**	4.6mA	4.6mA	5.4mA	N/A	N/A
Radio-only TX Power (@3v @0dBm)	**2.1 mA**	4.8mA	4.9mA	5.3mA	N/A	N/A
SoC Rx Power (@3V)	**1.4 mA**	N/A	N/A	N/A	4.6mA	3.6mA
SoC Tx Power (@3v @0dBm)	**2.5 mA**	N/A	N/A	N/A	4.9mA	4.1mA
RX Sensitivity (1Mbps)	-95dBm	-95dBm	-96dBm	-96dBm	-96dBm	-98.9dBm
TX output	**10dBm**	8dBm	8dBm	4dBm	10dBm	6dBm
CPU	64Mhz M33F	64MHz M4F	64MHz M4F	64MHz M4F	48MHz RISC	76.8MHz M33

Atmosic Advantage:

- Extreme Low Power
- Advanced Computation
- Advanced Security
- Energy Harvesting
- On-demand Wake Up
- MCU-less Sensing

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Atmosic ATM33 vs. Competition - continue

	Atmosic ATM33	NRF52840	NRF52833	NRF52832	TLSR8271	EFR32BG22
AoA, AoD	Yes	No	Yes	Yes	Yes	Yes
Voice Scaling	**Yes**	No	No	No	No	No
Voice Activity Detect	**Yes**	No	No	No	No	No
ARM TrustZone	**Yes**	No*	No	No	No	Yes
AES	**256(HW)**	128	128	128	128	256
NVM/SRAM	512KB/128KB	1MB/256KB	512KB/128KB	512KB/64KB	512KB/32KB	512KB/32KB
ROM	**64KB**	No	No	No	No	No
ADC	11b	12b	12b	12b	14b	16b
Mic	Digital	Digital	Digital	Digital	Analog, Digital	Digital
Voltage	1V-4.2V	1.7V - 5.5V	1.7V - 5.5V	1.7V - 3.6V	1.8V - 3.6V	1.71V - 3.8V
On Demand Wake Up	**Yes**	No	No	No	No	No
Energy Harvesting	**Yes**	No	No	No	No	No
Operating Temperature	-40C to 85C	-40C to 85C	-40C to 105C	-40C to 85C	-40C to 85C	-40C to 125C

*nRF52840 included ARM CryptoCell cryptographic accelerator, which only supply the crypto toolbox functionality as an IP block (e.g. HW accelerated cryptography and secure key derivation, for Secure Boot with Root-of-Trust). **nRF52840 does not provide full *Trusted Execution Environment* as described by ARM TrustZone.**

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Backup Slides - ARM M33F vs M4

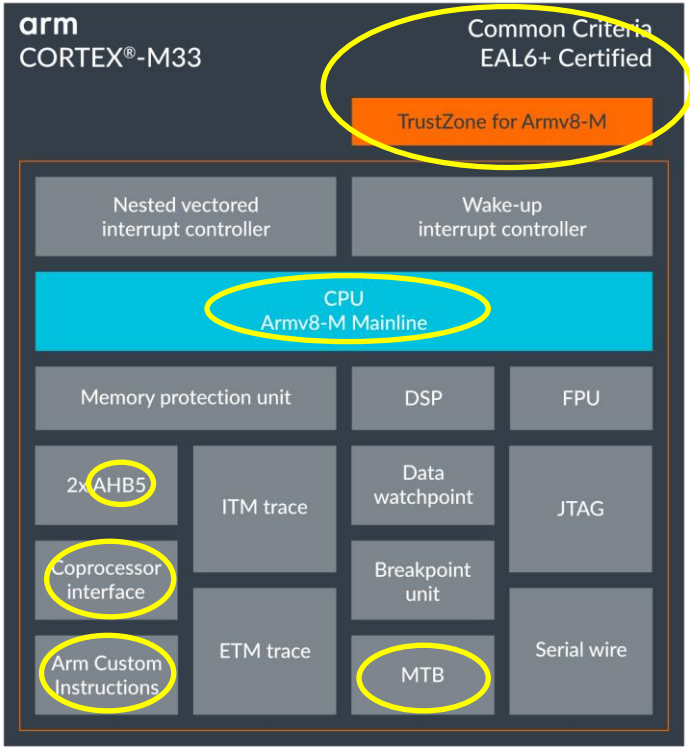
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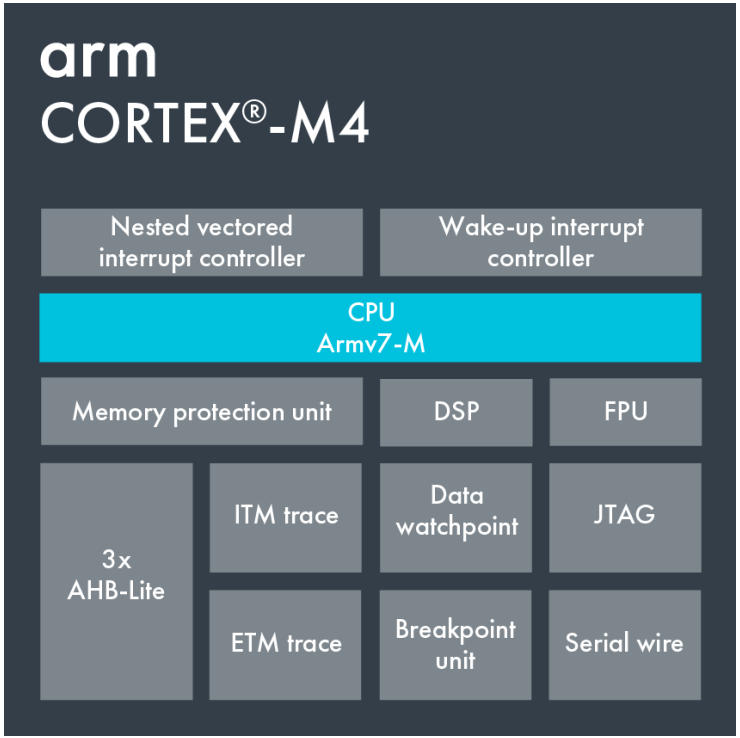
Summary - ARM M33F vs. M4 Summary

- The ATM33 Series is built upon the ARM® Cortex®-M33:
  - The first Armv8-M processor that supports ARM® TrustZone
  - Designed to address IoT markets that require highly efficient security and digital signal control.
- ARM M33F also offers:
  - Digital Signal Processing (DSP) extension
  - Floating-point Unit (FPU), to meet high precision calculation requirements in applications like CGM .
  - Arm Custom Instructions (ACI) to speed up specific operations.
- The above are the reasons why Atmosic has selected the new Arm Cortex-M33 over alternative cores, including the Cortex-M4, found in many competitor solutions.

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ARM M33F vs. M4 – Block Diagrams





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ARM M33F vs. M4 Feature Differences

Feature	M33F	M4	Benefits
ARM Architecture	ARMv8-M	ARMv7-M	ARMv8-M Supports ARM TrustZone
Advanced Microcontroller BUS Architecture (AMBA)	2x AMBA5 AHB	3x AMBA3 AHB-Lite	AMBA5 is the latest specification and offers high hardware coherency, enhanced transport layer efficiency, and congestion reduction
Security	ARM TrustZone	N/A	ARM TrustZone Secure Execution Environment
Accelerator support through coprocessor interface	Support up to 8	N/A	Enables the coprocessor to offload certain application-specific functions (currently not supported in ATM33)
Interrupts	Up to 480 physical interrupts	Up to 240 physical interrupts	Increased interrupt handling capacity/capability
Micro Trace Buffer (MTB)	supported	N/A	MTB is a peripheral used for instruction tracing. No special debuggers are needed to view the trace history. Useful for field products debugging.

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